Duc Le

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First-year mechanical engineering student at Rutgers with hands-on experience in CAD, FEA, and competition robotics.

EDUCATION

Rutgers University–New Brunswick *Expected May 2030*
B.S. Mechanical Engineering *GPA: N/A*
Relevant coursework: Statics, Multivariable Calculus, Engineering Economics, ID3EA I, AP Calculus I-II, AP Analytical Physics

ATLANTIC COUNTY INSTITUTE OF TECHNOLOGY *June 2026*
Academy of Math, Engineering, and Science *GPA: 4.2*

PROJECTS

FEA & GD&T *| Personal Project* *October 2026 - Present*
- *Applying FEA in Onshape to check whether the VEX robot's 3D-printed aligner plates survive impact loading*
- *Ran a preliminary load case (−200 lbf per plate) and examined stress around cutouts, holes, and spacers; validating constraints and material inputs next*
- *Planning a dimensioned drawing with GD&T callouts for the aligner's fits and tolerances*

Volumetric Display *| Lead Designer | Collaborative Project* *August 2026 - Present*
- *Collaborating on a swept-volume 3D display where a fast-spinning surface and synced LEDs form an image viewable from any angle; handling mechanical design in Onshape and SolidWorks*
- *Redesigned the rotating surface from a solid wall to 20 modular radial arms to reduce exposed area; packaged the motor, slip ring, and split printable shaft*
- *Running preliminary FEA on the arms (~23 psi at 0.28 lbf); dynamic validation at speed is next*

Quadruped Robot *| Lead Designer | Collaborative Project* *October 2026 - Present*
- *Leading mechanical design in Onshape for a folding four-legged robot; extended the chassis 160 → 225 mm and shortened the upper legs 143 → 135 mm to package the servos*
- *Iterated locating slots, servo-axis alignment, and mirrored printed supports for assembly*
- *Assembled four legs with one actuated; walking control is in development with the team's lead programmer*

Sprocket *| Team Leader & Lead Designer* *Sept 2025 – Apr 2026*
- *Led CAD design in Onshape and AutoCAD for a 4-person team, building a competition robot from scratch that solved prior teams' failures in weight, navigation drift, and documentation*
- *Designed the sensor mounts and tracking-wheel hardware for a closed-loop odometry and PID navigation system*
- *Won Judge's Award for engineering process and documentation; placed 3rd at SkillsUSA*

Launch Vehicle Project *| Lead Designer, EGR 101 (Embry-Riddle)* *Sept 2024 – May 2025*
- *Led design of a 125 ft two-stage orbital rocket concept for a 4-person team, sized for a 4,500 lb payload to low Earth orbit*
- *Ran structural calculations for max dynamic pressure, sizing a 1/16-inch Aluminum 6063-T5 skin and titanium matrix composite bulkheads, and built a spreadsheet to check applied stress against allowable*
- *Modeled staging and orbital trajectory in MATLAB and the vehicle in SolidWorks*

TECHNICAL SKILLS

CAD & Analysis: Onshape (FEA), SolidWorks, AutoCAD

Design & Fabrication: Mechanical Design, 3D Printing (FDM), Reverse Engineering

Programming & Tools: C++, Python, MATLAB, Excel / Google Sheets

Languages: English, Vietnamese, Chinese

HONORS & AWARDS

Top 10 Academic Student: Recognized for academic excellence at ACIT.